

Dr. Doğuş C. Öksüz

Curriculum vitae

Data Scientist at the Office for National Statistics

Data Science Campus

Emails: dogus.oksuz [at] ons [dot] gov [dot] uk

d.c.oksuz1[at] lancaster [dot] ac [dot] uk

Profile

I am a cognitive scientist and data scientist interested in how humans learn from the statistical structure of language. My work asks a simple but far-reaching question: what kind of computational machinery allows children and adults to extract patterns from linguistic experience and use them to learn, process, and represent language? In particular, my research focuses on:

- how monolingual and bilingual speakers process and represent words and multiword expressions;
- the role of frequency and distributional information in language learning and processing;
- individual differences in language learning and bilingual cognition; and
- how learning mechanisms adapt to the statistical properties of typologically diverse languages.

Methodologically, I combine experimental design, corpus analysis, statistical modelling, machine learning, and reproducible data-science workflows to analyse complex linguistic and behavioural datasets with rigour, transparency, and interpretability.

As an educator, I am committed to delivering high-quality teaching, designing materials that support students from diverse backgrounds, and enhancing the learning experiences of undergraduate and postgraduate students. My teaching focuses on making statistics, data science, research methods, and language science accessible, engaging, and practically useful.

Appointments

Data Scientist at the Office for National Statistics

Central Data Science and AI

2025-Present

Teaching Fellow at the University of Edinburgh

School of Mathematics

2024

Postdoctoral Researcher at Lancaster University Department of Linguistics and English Language	2023-2024
Postdoctoral Researcher at the University of Birmingham Department of Modern Languages	2023
Postdoctoral Researcher at the University of Cambridge Department of Theoretical and Applied Linguistics	2021-2022
Postdoctoral Researcher at the University of Leeds Department of Education	2021

Education

Ph.D. in Linguistics University of Lancaster, Department of Linguistics and English Language Thesis title: Collocational processing in typologically different languages: English and Turkish. Evidence from corpora and psycholinguistic experimentation Supervisors: Professors Patrick Rebuschat and Vaclav Brezina	2019
M.S.c. in Language Education (awarded with honours) University of Edinburgh, Moray House School of Education	2014
B.A. in Linguistics & Foreign Language Education Boğaziçi University, Foreign Language Education	2013

Publications

In Preparation

Ott, C., Rebuschat, P., **Öksüz, D.**, Bottini, R. Distributional Predictors of Lexical Bundle Processing: Effects of Frequency, Association Strength, and Length

Öksüz, D., Cindy Ge, Göymen D., Rebuschat, P., & Christiansen M. Chunking Ability Predicts Collocational Processing in Heritage and Non-Heritage Turkish Speakers

Manuscripts under Review / Revision

Öksüz, D., Akef S., Mendes, A., Rebuschat, P. Noun-adjective collocations in L2 Portuguese: Exploring the effects of proficiency and native language in a large-scale learner corpus [[Open data & Code](#)] [Preprint]

Refereed Journal Articles

Öksüz, D., Alexopoulou, T., Derkach, K., & Tsimpli, I. M. (2025). The influence of L1 typology on the acquisition of the L2 English article: A large-scale corpus study. *Second Language Research*. [\[Open data & Code\]](#) [\[Preprint\]](#)
<https://doi.org/10.1177/02676583251395876>

Öksüz, D., Brezina, V. Monaghan, P., & Rebuschat, P. (2024). Individual word and phrase frequency effects in collocational processing: Evidence from typologically different languages, English and Turkish. *Journal of Experimental Psychology: Learning, Memory, and Cognition*. 50(8), 1287-1314. [\[Open data & Code\]](#) <https://doi.org/10.1037/xlm0001324>

Öksüz, D., Brezina, V. and Rebuschat, P. (2021), Collocational processing in L1 and L2: The effects of word frequency, collocational frequency, and association. *Language Learning*, 71: 55-98. [\[Open data & Code\]](#)
<https://doi.org/10.1111/lang.12427>

Oral and Poster Presentation

Öksüz D., Akef S., Mendes, A., Rebuschat, P. (2023, September). Tracking the development of multiword expressions in L2 Portuguese: A learner corpus analysis. Presented at EuroSLA 32. University of Birmingham, Birmingham.

Öksüz, D., Alexopoulou, T., Derkach, K., & Ianthi I. M. (2022, August). L1 and L2 Typological Distance Effects on the Learnability of Articles in L2 English. Presented at EuroSLA 31. University of Fribourg, Fribourg.

Öksüz, D., Rebuschat, P., Brezina, V., & Monaghan P. (2022, June). Individual word and phrasal frequency effects in collocational processing: Evidence from typologically different languages, English and Turkish. Presented at Interdisciplinary Advances in Statistical Learning Conference. Basque Center on Cognition, Brain, and Language, San Sebastian, Spain.

Öksüz, D., Brezina, V., & Rebuschat, P. (2021, September). Individual word and phrasal frequency effects in collocational processing: Evidence from typologically different languages, English and Turkish. Presented at X-PPL. University of Zurich, Zurich.

Selected Invited Talks

Öksüz, D. (2025, June). Processing and Using Collocations Across Populations and Languages: Insights from Experimental and Learner Corpus Approaches. Brussels Centre for Language Studies, Vrije Universiteit Brussel.

Öksüz, D. (2024, December). Chunking ability predicts collocational processing in heritage speakers. Second Language Learning and Teaching Research Group, Lancaster University.

Öksüz, D. (2023, Feb). Collocational processing: The effects of word frequency, phrasal frequency and linguistic typology. Laboratory of Phonetics, University of Glasgow.

Öksüz, D. (2021, March). Collocational processing in L1 and L2: The effects of word frequency, collocational frequency, and association: Combining Corpus and Experimental Approaches. Psycholinguistics Coffee Research Group, University of Edinburgh.

Öksüz, D. (2021, January). Collocational processing in L1 and L2: The effects of word frequency, collocational frequency, and association: Combining Corpus and Experimental Approaches. The Cambridge Processing and Acquisition of Language Lab, University of Cambridge.

Funding

Individual Differences in Turkish Heritage Speakers' Grammatical Processing **2023-2024**

Funder: Lancaster University FASS Research Fund (Co-I), Project Partner: Professor Patrick Rebuschat (PI)

Awards

2022: Top Cited Article Award: Language Learning. Öksüz et al., 2021

Teaching

- University of Edinburgh, Sep–Dec 2024
Delivered seminars and workshops for **Introduction to Data Science** (UG) and **Generalised Regression Models** (PG) in the School of Mathematics. Focused on exploratory data analysis, visualization, statistical modelling, and prediction. Provided personalized feedback on assignments and projects and held regular office hours for academic and pastoral support.

- St Andrews College Language Schools, Jul–Aug 2024
Delivered a summer language program for international students (ages 12–17). Designed and taught engaging, level-appropriate lessons to promote language acquisition and communicative confidence. Facilitated interactive activities and offered individualized support to enhance learning outcomes.
- *University of Cambridge, Oct 2022 – Jun 2023*
Delivered undergraduate seminars on the undergraduate course **Language, Brain & Society**, focusing on various topics in Sociolinguistics, First and Second Language Acquisition, and Psycholinguistics. Encouraged critical thinking and interdisciplinary engagement with course material. Provided regular, personalized feedback and held office hours for academic and pastoral support.
- *Lancaster University, Oct 2018 – Jan 2019*
Taught third-year course **Corpus-based English Language Studies** seminars and lectures on corpus research, including a lecture on Corpus Linguistics and Language Acquisition. Provided academic and pastoral support through office hours and marked assignments.

Research Skills

- Advanced proficiency in **R** for statistical analysis, data visualisation, and modelling
- Advanced expertise in **multilevel modelling** and advanced quantitative methods
- Proficiency in **Python** for data analysis, data wrangling, NLP, and reproducible research workflows,
- Experience with machine learning methods and relevant Python libraries, including **pandas**, **NumPy**, **scikit-learn**, and **seaborn**
- Experience designing and implementing behavioural experiments using **PsychoPy**, and **Gorilla**, including online deployment
- Experience with NLP and corpus-analysis tools, including **TreeTagger**, **Sketch Engine**, and, **#LancsBox**,

Professional Memberships

- Member of **Bilingualism Matters**, an international research and information network on bilingualism and language learning.